Savera Ansari

Assignment 1

(NLP)

Text Classifier for Sarcasm Detection in Urdu Text



FAST National University of Computer and Emerging Sciences

Submitted To: Dr. Muhammad Rafi

Submitted By: Savera Ansari
ID: k25-8009

1. Task: The task is binary classification of Urdu social-media posts: sarcastic (1) versus non-sarcastic (0).

2. Dataset: The task is binary classification of Urdu social-media posts: sarcastic (1) versus non-sarcastic (0). The uploaded notebook loads 20,060 CSV rows and, after removing missing/invalid entries and deduplicating exact text, retains 15,126 unique labelled examples (8,676 non-sarcastic; 6,450 sarcastic). Stratified train/validation/test splits contain 10,588 / 2,269 / 2,269 examples (70% / 15% / 15%)

3. Exploratory Analysis: EDA checks class balance, missing values, duplication, text lengths and sample posts. Social-media features include Urdu spelling variation, Urdu–Punjabi mixtures, Roman/English tokens, mentions, hashtags, URLs and emojis. In the source data, laughing emojis occur in 44.1% of sarcastic versus 30.0% of non-sarcastic rows, while any emoji occurs in 77.8% versus 88.1%, respectively. These differences motivate controlled tests rather than assuming every emoji signals sarcasm.

4. Data preparation: Cleaning removes empty/missing examples and explicit label-revealing tags (e.g., #sarcasm); repeated texts are grouped, and label conflicts are tracked using soft labels. Urdu-aware preprocessing normalizes Arabic/Urdu character variants, diacritics and whitespace while preserving informative stylistic signals. Raw text, normalized text, emoji-removed text and emoji-as-text variants are retained for ablation. Exact duplicate removal precedes splitting to reduce direct train–test leakage.

5. Representations: Features include TF-IDF word and character n-grams; additional manually engineered Urdu sarcasm phrases, negations, political terms, lexical polarity incongruity, punctuation and emoji indicators. The neural experiment separates word tokens from emoji tokens. These choices reflect short, informal and code-mixed Urdu posts, where both word meaning and expressive style contribute to irony.

6. Trained models and experimental design: Classical: multinomial Naive Bayes and tuned Logistic Regression / linear SVM with word, character and combined TF-IDF representations. Neural: single-branch comparison and two-branch BiLSTM (word embeddings + attention, separate emoji branch); Urdu fastText vectors matched 14,726 of 17,925 eligible unique tokens (82.2%). Transformer: fine-tuned cardiffnlp/twitter-xlm-roberta-base with single-input and text–emoji pair encoding. Ensembles average predicted probabilities; weights were selected on validation (two-model transformer/TF-IDF 0.55/0.45; three-model transformer/TF-IDF/BiLSTM 0.60/0.20/0.20).
The notebook sets a fixed seed and compares models primarily by macro-F1. The final held-out test table is produced after the model-selection stages. Accuracy, per-class precision/recall/F1 and confusion matrices are used for interpretation.

7. Results and Ablation:

7(a). Final held-out test performance

Macro-F1 is the primary ranking metric. Class 0 is non-sarcastic, class 1 sarcastic. The following figures reproduce the notebook's saved final test outputs.

Model	Accuracy	P_0 / R_0 / $F1_0$	P_1 / R_1 / $F1_1$	Macro-F1
Three-model ensemble	0.8435	0.8719 / 0.8524 / 0.8620	0.8074 / 0.8316 / 0.8193	0.8407
Transformer + TF-IDF	0.8396	0.8646 / 0.8540 / 0.8592	0.8069 / 0.8202 / 0.8135	0.8364
Twitter XLM-R (pair)	0.8308	0.8767 / 0.8201 / 0.8475	0.7776 / 0.8450 / 0.8099	0.8287
Twitter XLM-R (single)	0.8299	0.8675 / 0.8301 / 0.8484	0.7842 / 0.8295 / 0.8062	0.8273
TF-IDF word+char SVM	0.8219	0.8262 / 0.8732 / 0.8490	0.8154 / 0.7531 / 0.7830	0.8160
TF-IDF word+char LR	0.8206	0.8234 / 0.8747 / 0.8483	0.8162 / 0.7479 / 0.7806	0.8144
BiLSTM two-branch	0.8189	0.8306 / 0.8593 / 0.8447	0.8017 / 0.7645 / 0.7827	0.8137
TF-IDF+handcrafted LR	0.8149	0.8227 / 0.8632 / 0.8425	0.8031 / 0.7500 / 0.7756	0.8091

7(b). Confusion matrices (test)

Rows = true class; columns = predicted class. Counts are reconstructed from the notebook's rounded recall values and known test class sizes; verify against freshly exported raw confusion matrices before final submission.

Model	True 0 → Pred 0	True 0 → Pred 1	True 1 → Pred 0	True 1 → Pred 1
Three-model ensemble	1109	192	163	805
TF-IDF word+char LR	1138	163	244	724

The ensemble correctly detects more sarcastic posts than the TF-IDF LR baseline (805 versus 724) while sacrificing some non-sarcastic specificity (1,109 versus 1,138 true negatives). This trade-off explains the ensemble's improved macro-F1.

8. Validation Ablation findings

Comparison	Macro-F1 (validation)	Interpretation
Emoji kept / removed	0.8144 / 0.8063	Emojis offer useful information
Emoji-to-text	0.8159	Small gain over emoji kept
Raw / normalized	0.8168 / 0.8144	Normalization not universally beneficial
Word / character / both	0.8128 / 0.8138 / 0.8144	Combined features slightly stronger
No / all handcrafted	0.8144 / 0.8191	All engineered features help modestly
Tags removed / kept	0.8144 / 0.8183	Label-revealing tags inflate scores
Deduplicated / row-split duplicates	0.8144 / 0.8209	Leaky duplicate split overstates performance

A5 and A6 are deliberately leaky diagnostic demonstrations, not valid performance baselines. The row-split duplicate demonstration also reports 33.6% train/test exact-text overlap.

9. Linguistic error analysis

For the transformer + TF-IDF ensemble analyzed in the notebook, 364 of 2,269 test posts (16.0%) were misclassified. Error rates are higher for conflicting/ambiguous labels (28.4%), Punjabi/regional expressions (25.0%) and laughing-emoji posts (21.7%). Below are five real notebook error cases; interpretations are qualitative possibilities, not verified causal explanations.

FP — یہ سب جالی اکاونٹ ہے اور ن لیگ ان کی موجد ہے 😄

Political criticism and the smiling emoji may resemble mocking discourse even though the annotation is non-sarcastic.

FP — لکھ لعنت وی تیری پلاسٹک دی بوتھی تے۔ 😂😂

Punjabi insult with laughter can blur ridicule, anger and sarcasm; token/context cues may be over-weighted.

FP — زرائع سے معلوم ہورہا ہے سندھ پولس نے دو طیارے مار گرائے ہیں اور ایک ونگ کمانڈر بھی پکڑ لیا ہے 😭

An implausible-sounding claim plus a sad emoji could trigger irony heuristics despite a non-sarcastic label.

FN — ارے پوچھنے بتانے کی کیا ضروت ھے سب کو پتا ھے کہ کل رات تم نے کہاں اور کس کے ساتھ گزاری ھے 😂😂😂😂

The post is annotated sarcastic, but its conversational teasing needs interpersonal context; conflicting original labels were recorded.

FN — خان_گروپ 👈

An extremely short hashtag-like expression gives little semantic context; the model cannot reliably infer the annotated sarcasm.

10. Limitations and conclusions

The three-model ensemble achieves the highest reported test macro-F1 (0.8407), ahead of the single Twitter XLM-R (0.8273) and TF-IDF word+character LR (0.8144). Nevertheless, observed gains come from one recorded split; different seeds and broader validation could change rankings. Sarcasm remains difficult when political background, local idiom, emoji intent and conversation context are absent.

Caution: the notebook's existing ablation discussion still quotes results from an older run (~0.717); it should be updated to the 0.81–0.82 validation values above. Its qualitative-error table is also not fully filled with manual explanations. These report commentaries are initial case reviews and should be checked by the student against the corresponding messages and labels.

11. Reproducibility and acknowledgments

Notebook: 25k_8009_NLP1.ipynb. Saved outputs include model-comparison/ablation CSVs, test predictions, TF-IDF vectorizers/classifier, BiLSTM weights, transformer weights/tokenizer and config. Reproducing results requires the same dataset, preprocessing, pretrained model checkpoints, seed and dependency versions.